Distributional Preference Alignment of LLMs via Optimal Transport

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Outline

Pointwise Preference Optimization

From Pointwise to Distributional Preference

Distributional Preference Relaxation to a 1D Optimal Transport Problem

Statistical Properties

Results

The Alignment Problem

- Find a policy that maximizes a reward r while staying close to a reference policy π_{ref}:

$$\max_{\pi_\theta} \int r d\pi_\theta - \beta \text{KL}(\pi_\theta || \pi_{\text{ref}})$$

- The optimal policy is given by:

$$\pi_\theta(y|x) = \frac{\pi_{\text{ref}}(y|x) \exp(\frac{r(x,y)}{\beta})}{Z(x)}$$

DPO as a Pointwise Preference Approach 1/2

Direct Preference Optimization (DPO)
[Rafailov et al.(2024)Rafailov, Sharma, Mitchell, Manning, Ermon, and Finn]

- The reward optimized by the LLM:

$$r_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log(Z(x))$$

- $Z(x)$ is a normalization constant.
- Paired preference dataset: $(X, Y_+, Y_-) \sim \mu$
  - Y_+: positive (chosen) response
  - Y_-: negative (rejected) response

DPO as a Pointwise Preference Approach 2/2

- Objective: Minimize the logarithmic sigmoid loss:

$$\min_{\theta \in \Theta} -\mathbb{E}_{(x,y_+,y_-)\sim\mu} \log(\sigma(\beta(r_\theta(x,y_+) - r_\theta(x,y_-))))$$

- Simplified form (normalization $Z(x)$ cancels out):

$$\min_{\theta} -\mathbb{E}_{(x,y_+,y_-)\sim\mu} \log\left(\sigma\left(\beta \log\left(\frac{\pi_\theta(y_+|x)}{\pi_{\text{ref}}(y_+|x)}\right) - \beta \log\left(\frac{\pi_\theta(y_-|x)}{\pi_{\text{ref}}(y_-|x)}\right)\right)\right)$$

- Interpretation as pointwise preference:

$$\boxed{\log\left(\frac{\pi_\theta(y_+|x)}{\pi_{\text{ref}}(y_+|x)}\right) \geq \log\left(\frac{\pi_\theta(y_-|x)}{\pi_{\text{ref}}(y_-|x)}\right), \quad \forall (x, y_+, y_-) \sim \mu.} \tag{1}$$

- Relaxation through logistic loss, suggests other algorithms (e.g., hinge loss in SLIC [Zhao et al.(2023)Zhao, Khalman, Joshi, Narayan, Saleh, and Liu]).

Pointwise Preference

- Pointwise Preference positive versus negative:

$$\boxed{\log\left(\frac{\pi_\theta(y_+|x)}{\pi_{\text{ref}}(y_+|x)}\right) \geq \log\left(\frac{\pi_\theta(y_-|x)}{\pi_{\text{ref}}(y_-|x)}\right), \quad \forall(x, y_+, y_-) \sim \mu.} \tag{2}$$

- Equivalently: pointwise Preference of margin (positive - negative) of Model versus Reference:

$$\boxed{\log\frac{\pi_\theta(y_+|x)}{\pi_\theta(y_-|x)} \geq \log\frac{\pi_{\text{ref}}(y_+|x)}{\pi_{\text{ref}}(y_-|x)}, \quad \forall(x, y_+, y_-) \sim \mu.} \tag{3}$$

Outline

Pointwise Preference Optimization

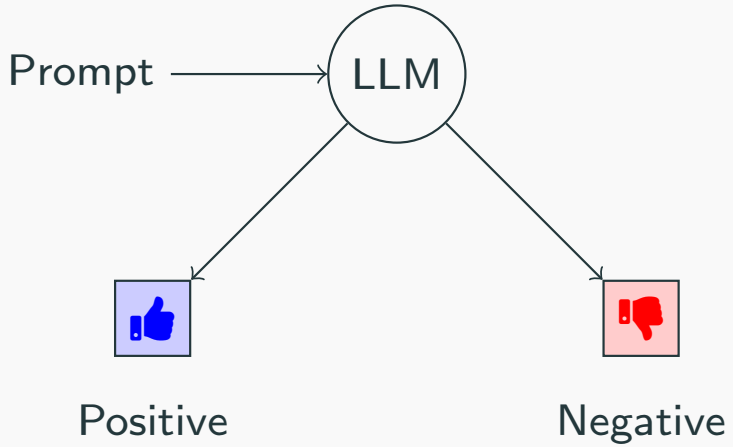
From Pointwise to Distributional Preference

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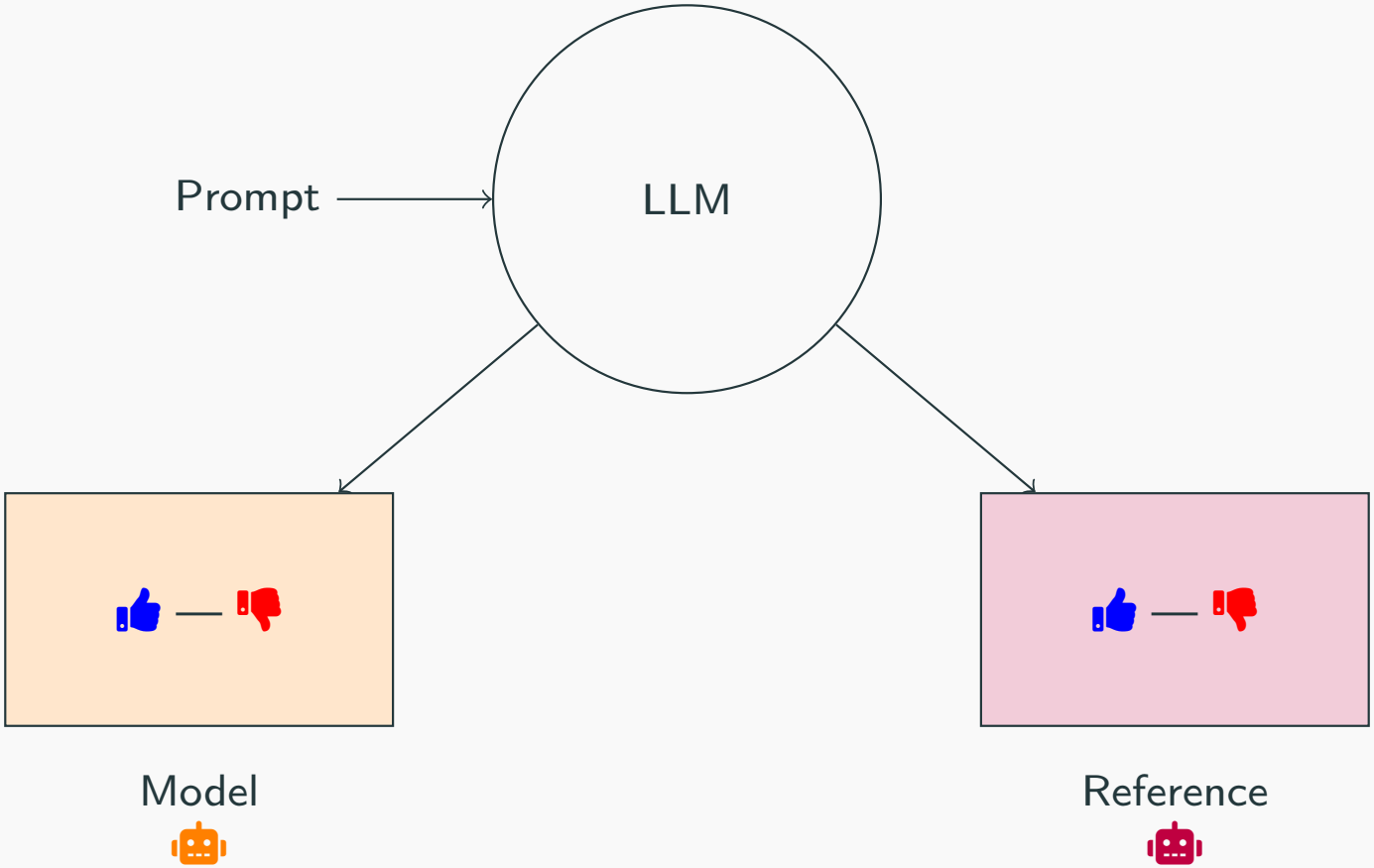
Results

Pointwise Preference (Positive versus Negative)



$$\log\left(\frac{\pi_\theta(y_+|x)}{\pi_{\text{ref}}(y_+|x)}\right) \geq \log\left(\frac{\pi_\theta(y_-|x)}{\pi_{\text{ref}}(y_-|x)}\right)$$

Pointwise Preference, Model Versus Reference based on Margins



$$\log \frac{\pi_\theta(y_+|x)}{\pi_\theta(y_-|x)} \geq \log \frac{\pi_{\text{ref}}(y_+|x)}{\pi_{\text{ref}}(y_-|x)}$$

Partial Order on One Dimensional Distribution: First Order Stochastic Dominance

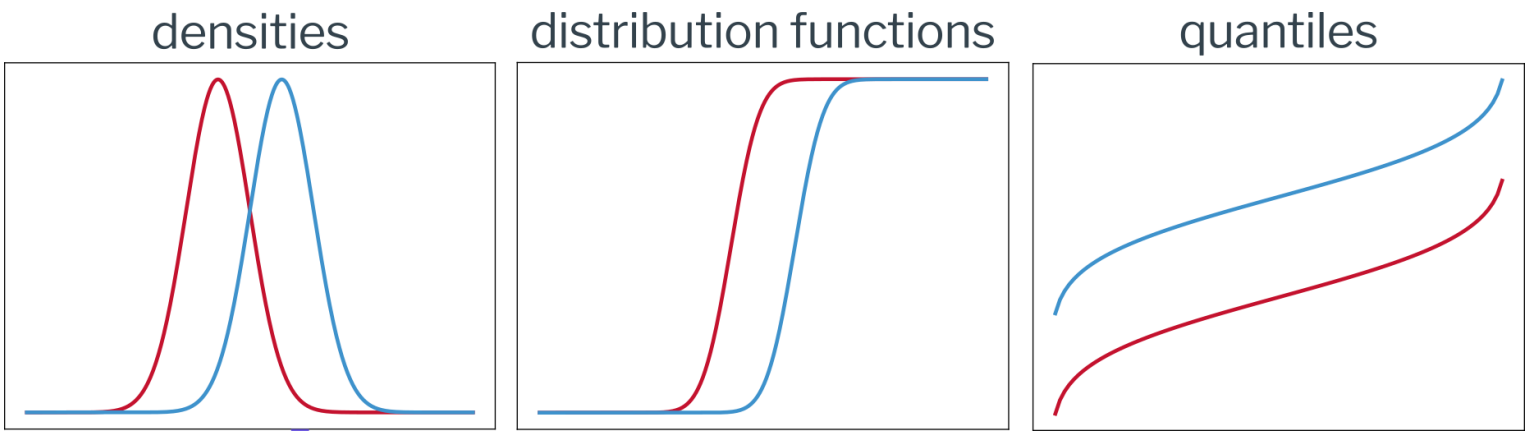
Definition:

- For a real random variable Z, the left-continuous inverse of the Cumulative Distribution Function (CDF) F_Z is denoted by $F_Z^{(-1)} : [0,1] \to \overline{\mathbb{R}}$.
- This inverse, also known as the quantile function $Q_Z(p)$, is defined as:
$$Q_Z(p) = F_Z^{(-1)}(p) = \inf\{\eta : F_Z(\eta) \geq p\} \text{ for } p \in [0,1].$$

Definition (First Order Stochastic Dominance (FSD):)

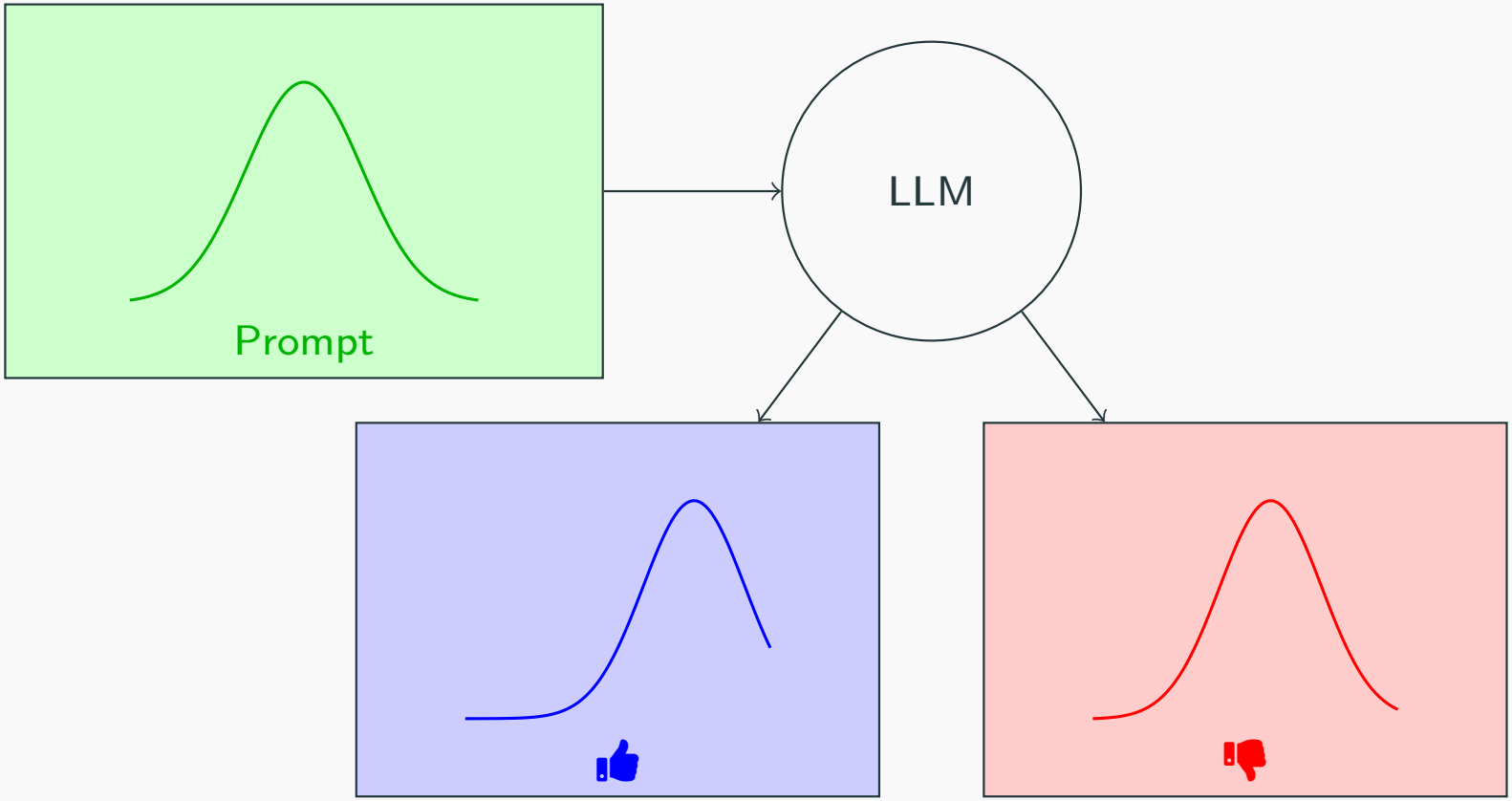
Given two random variables Z_1 and Z_2, Z_1 is said to dominate Z_2 in the first order if Z_1 has larger quantiles than Z_2 for all percentiles p:
$$Z_1 \underset{\text{FSD}}{\succcurlyeq} Z_2 \iff Q_{Z_1}(p) \geq Q_{Z_2}(p), \quad \forall p \in [0,1].$$

Stochastic Dominance and OT



$X \underset{\text{FSD}}{\succcurlyeq} Y$ equivalent to $F_Y^{-1}(t) \leq F_X^{-1}(t)\ \forall t \in (0,1)$ so

Distributional Preference Unpaired (based on rewards of positive and negatives)

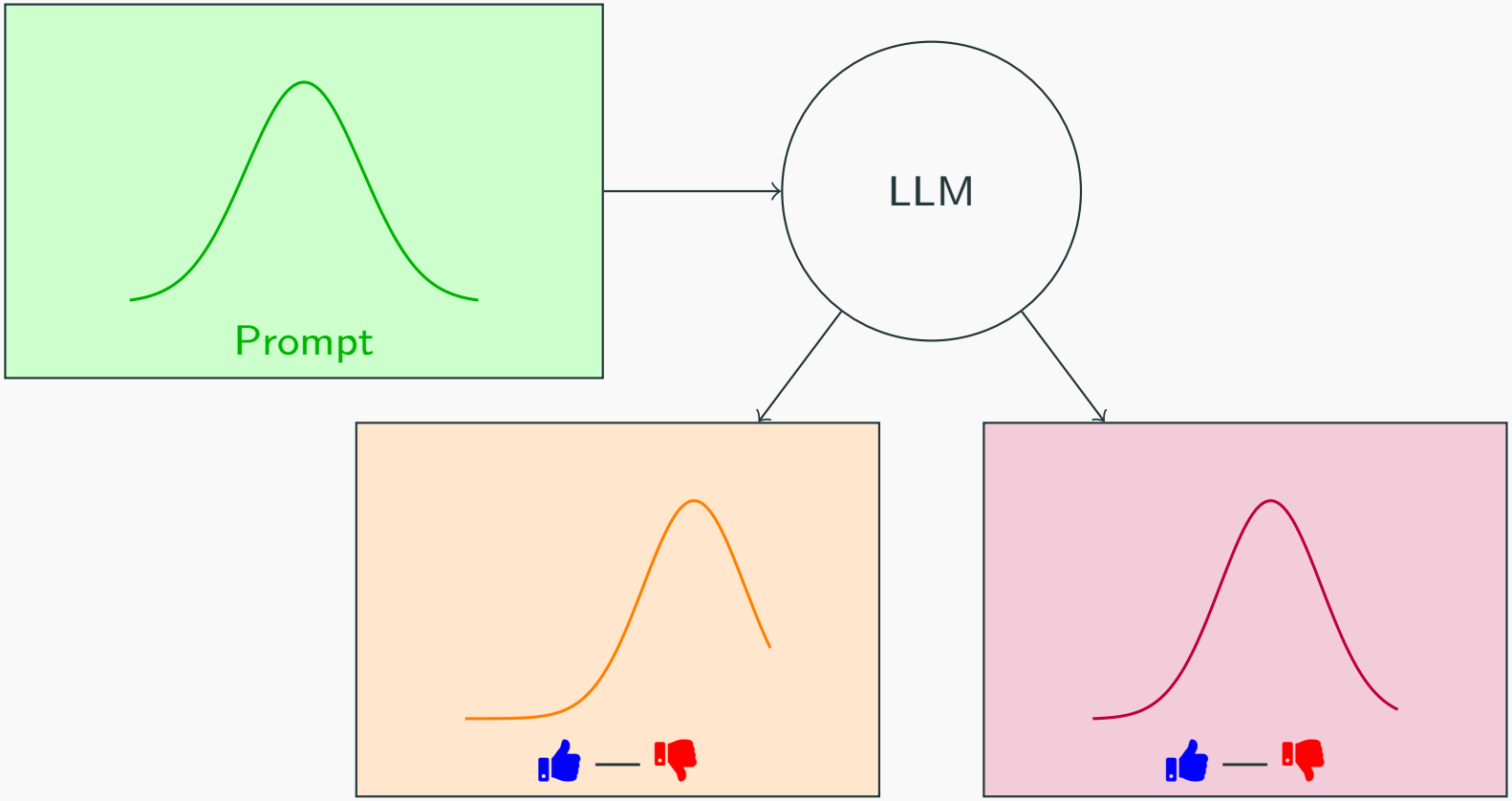


Positive Negative

$$\log \frac{\pi_\theta(Y_+|X_+)}{\pi_{\text{ref}}(Y_+|X_+)} \underset{\text{FSD}}{\succcurlyeq} \log \frac{\pi_\theta(Y_-|X_-)}{\pi_{\text{ref}}(Y_-|X_-)}.$$

does not need paired positive and negative answer for each prompt!

Distributional Preference Paired
(based on margins of rewards between Model and Reference)



$$\log \frac{\pi_\theta(Y_+|X)}{\pi_\theta(Y_-|X)} \succcurlyeq_{\text{FSD}} \log \frac{\pi_{\text{ref}}(Y_+|X)}{\pi_{\text{ref}}(Y_-|X)}$$

needs paired positive and negative answer for each prompt!

Distributional Unpaired Preference

- No access to triplets of prompts and positive/negative responses (x, y_+, y_-).
- Separate access to:
  - $\mu_+ \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$: Distribution of positive prompt/response pairs (X_+, Y_+) to be highly rewarded.
  - $\mu_- \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$: Distribution of negative samples (X_-, Y_-) to be associated with low reward.
- Define the distributional preference as follows:

Definition (Distributional Preference in the Unpaired Setting)

A policy π prefers distributionally μ_+ over μ_- with respect to a reference policy π_{ref} if:

$$\log \frac{\pi_\theta(Y_+|X_+)}{\pi_{\text{ref}}(Y_+|X_+)} \underset{\text{FSD}}{\succcurlyeq} \log \frac{\pi_\theta(Y_-|X_-)}{\pi_{\text{ref}}(Y_-|X_-)}.$$

In other words, noting $r_u \circ \pi_\theta(x, y) = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$, the distributional preference in the unpaired setting means that we have the following constraint:

$$(r_u \circ \pi_\theta)_\sharp \mu_+ \underset{\text{FSD}}{\succcurlyeq} (r_u \circ \pi_\theta)_\sharp \mu_-. \tag{4}$$

Distributional Paired Preference

We define below more formally the paired distributional preference via stochastic dominance:

Definition (Distributional Preference in the Paired Setting)

We say that the policy π_θ distributionally dominates π_{ref} in terms of log probability ratio of positive and negative responses if:

$$\log \frac{\pi_\theta(Y_+|X)}{\pi_\theta(Y_-|X)} \underset{\text{FSD}}{\succcurlyeq} \log \frac{\pi_{\text{ref}}(Y_+|X)}{\pi_{\text{ref}}(Y_-|X)}.$$

Noting $r_p \circ \pi_\theta(x, y_+, y_-) = \log \frac{\pi_\theta(y_+|x)}{\pi_\theta(y_-|x)}$ this can be written as follows:

$$(r_p \circ \pi_\theta)_\sharp \mu \underset{\text{FSD}}{\succcurlyeq} (r_p \circ \pi_{\text{ref}})_\sharp \mu. \tag{5}$$

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Learning Policies with Distributional Constraints

- FSD unpaired Constraint:

  Find $\pi_\theta \in \mathcal{H}$ such that $(r_u \circ \pi_\theta)_\sharp \mu_+ \underset{\text{FSD}}{\succcurlyeq} (r_u \circ \pi_\theta)_\sharp \mu_-$ (FSD unpaired)

- FSD Paired Constraint:

  Find $\pi_\theta \in \mathcal{H}$ such that $(r_p \circ \pi_\theta)_\sharp \mu \underset{\text{FSD}}{\succcurlyeq} (r_p \circ \pi_{\text{ref}})_\sharp \mu$ (FSD paired)

- Both Problem can be abstracted out to:

  Find $\theta \in \Theta$ such that : $U_\theta \underset{\text{FSD}}{\succcurlyeq} V_\theta$

A Convex Relaxation of FSD

$$\boxed{U_\theta \underset{\text{FSD}}{\succcurlyeq} V_\theta \iff Q_{U_\theta}(t) \geq Q_{V_\theta}(t), \forall t \in [0,1].}$$

We can relax this problem to minimizing the violation of the FSD order:

$$\boxed{\min_{\theta \in \Theta} \varepsilon(\theta) := \int_0^1 h(Q_{U_\theta}(t) - Q_{V_\theta}(t))dt,}$$

where h penalizes the violation of dominance of U_θ on V_θ

- **Indicator Loss**: $h(x) = \mathbb{1}_{x<0}$.
- β-**squared Hinge Loss**: For a margin $\beta > 0$, $h(x) = (\beta - x)_+^2$.
- β-**logistic Loss**: $h(x) = \log(1 + \exp(-\beta x))$.
- β-**Least Squares**:
  - Although not a convex relaxation of the 0/1 loss, the least squares loss has been used in classification .
  - In the context of alignment, it was used in IPO [Azar et al.(2024)Azar, Guo, Piot, Munos, Rowland, Valko, and Calandriello].
  - $h(x) = (\beta - x)^2$.

Quantile Violation as an Optimal Transport Problem

Theorem (Theorem 2.9 and Proposition 2.17 in [Santambrogio(2015)])

Let $h : \mathbb{R} \to \mathbb{R}^+$ be a convex function we have for two real random variables U, V, with measures μ_U, μ_V:

$$\int_0^1 h(Q_U(t) - Q_V(t))dt = \min_{\gamma \in \Pi(\mu_U, \mu_V)} \int h(u - v)d\gamma(u, v) = \mathrm{OT}_h(\mu_U, \mu_V)$$

and $\gamma^* = (Q_U, Q_V)_\sharp \mathcal{L}_1([0, 1])$ is a minimizer (where $\mathcal{L}_1$ is the Lebesgue measure on $[0, 1]$). If furthermore h is strictly convex γ^* is the unique minimizer.

$$\min_{\theta \in \Theta} \int_0^1 h(Q_{U_\theta}(t) - Q_{V_\theta}(t))dt = \min_{\theta \in \Theta} \mathrm{OT}_h(\mu_{U_\theta}, \mu_{V_\theta}) = \min_{\theta \in \Theta} \min_{\gamma \in \Pi(\mu_{U_\theta}, \mu_{V_\theta})} \int h(u - v)d\gamma(u, v)$$

Optimizing Relaxed FSD is an inner 1D Optimal Transport problem with smooth and convex cost

Computational Algorithm via Sorting

- We consider empirical measures and solve for a fixed θ. We simplify notation by omitting θ.
- $\mathrm{OT}_h(\hat{\mu}_U, \hat{\mu}_V)$ is of interest, where:

$$\hat{\mu}_U = \frac{1}{n}\sum_{i=1}^{n}\delta_{u_i}, \quad \hat{\mu}_V = \frac{1}{n}\sum_{i=1}^{n}\delta_{v_i}.$$

- Due to the convexity of h , $\mathrm{OT}_h(\hat{\mu}_{U_\theta}, \hat{\mu}_{V_\theta})$'s optimal coupling is given by the north-west corner solution [Peyré and Cuturi(2019)] (Chapter 3, Section 3.4.2), informally matching i-th smallest elements of U and V.
- Formally, if we sort u_i and v_i into order statistics ($u^{(1)} \leq \cdots \leq u^{(n)}$, $v^{(1)} \leq \cdots \leq v^{(n)}$):

$$\mathrm{OT}_h(\hat{\mu}_U, \hat{\mu}_V) = \frac{1}{n}\sum_{i=1}^{n} h(u^{(i)} - v^{(i)}).$$

- Given empirical samples $\hat{\mu}_{U_\theta} = \frac{1}{n}\sum_{i=1}^{n}\delta_{u_\theta^i}$ and $\hat{\mu}_{V_\theta} = \frac{1}{n}\sum_{i=1}^{n}\delta_{v_\theta^i}$, with $u_\theta^{(i)}, v_\theta^{(i)}$ as order statistics:

AOT

$$\min_{\theta\in\Theta}\mathrm{OT}_h(\hat{\mu}_{U_\theta}, \hat{\mu}_{V_\theta}) = \min_{\theta\in\Theta}\frac{1}{n}\sum_{i=1}^{n} h(u_\theta^{(i)} - v_\theta^{(i)}) \quad \text{(AOT)}$$

AOT: Alignment Via Optimal Transport

Algorithm 1 AOT Unpaired

Input: π_θ, $\pi_{\text{ref}}, \beta > 0, \varepsilon > 0$,
Unpaired Preference Data: "Chosen" $\hat{\mu}_+^n$ and "Rejected" $\hat{\mu}_-^n$.
for iter $\leftarrow 1, n_{\text{iter}}$ **do**
  Get a Positive/Negative mini-batch
  $\{(x_{i,+}, y_{i,+}) \sim \hat{\mu}_+^n, i = 1 \dots b\}$
  $\{(x_{i,-}, y_{i,-}) \sim \hat{\mu}_-^n, i = 1 \dots b\}$
  Compute Rewards
  $u_\theta^i = \log \frac{\pi_\theta(y_{i,+}|x_{i,+})}{\pi_{\text{ref}}(y_{i,+}|x_{i,+})}$
  $v_\theta^i = \log \frac{\pi_\theta(y_{i,-}|x_{i,-})}{\pi_{\text{ref}}(y_{i,-}|x_{i,-})}$
  Sort Rewards
  $(u^{(1)} \dots u^{(b)}) = \text{Sort}(u_\theta^i)$
  $(v^{(1)} \dots v^{(b)}) = \text{Sort}(v_\theta^i)$
  Compute AOT logistic loss
  $\ell_\theta = -\frac{1}{b} \sum_{i=1}^b \log \sigma(\beta(u_\theta^{(i)} - v_\theta^{(i)}))$
  Update θ **with** PagedAdamw32bit
end for
Return π_θ

Algorithm 2 AOT Paired

Input: π_θ, $\pi_{\text{ref}}, \beta > 0$,
Paired Preference Data: $\hat{\mu}^n = \frac{1}{n} \sum_{i=1}^n \delta_{(x_i, y_{i,+}, y_{i,-})}$
for iter $\leftarrow 1, n_{\text{iter}}$ **do**
  Get a Positive/Negative mini-batch
  $\{(x_i, y_{i,+}, y_{i,-}) \sim \hat{\mu}^n, i = 1 \dots b\}$
  Compute Margins for $i = 1 \dots b$
  $u_\theta^i = \log \frac{\pi_\theta(y_{i,+}|x_i)}{\pi_\theta(y_{i,-}|x_i)}$
  $v_\theta^i = \log \frac{\pi_{\text{ref}}(y_{i,+}|x_i)}{\pi_{\text{ref}}(y_{i,-}|x_i)}$
  Sort Margins
  $(u^{(1)} \dots u^{(b)}) = \text{Sort}(u_\theta^i)$
  $(v^{(1)} \dots v^{(b)}) = \text{Sort}(v_\theta^i)$
  Compute AOT logistic loss
  $\ell_\theta = -\frac{1}{b} \sum_{i=1}^b \log \sigma(\beta(u_\theta^{(i)} - v_\theta^{(i)}))$
  θ **with** PagedAdamw32bit
end for
Return π_θ

Code in TRL on HuggingFace

```
import torch
import torchsort
```

```
def dpo_loss( ...
...
elif self.loss_type == "AOT_unpair":
    chosen_logratios = policy_chosen_logps - reference_chosen_logps
    rejected_logratios = policy_rejected_logps - reference_rejected_logps
    if self.sort_type == "hard_sort":
        chosen_logratios_sorted, _ = torch.sort(chosen_logratios, dim=0)
        rejected_logratios_sorted, _ = torch.sort(rejected_logratios, dim=0)
    elif self.sort_type == "soft_sort":
        chosen_logratios_sorted = torchsort.soft_sort(chosen_logratios, regularization_strength
            =0.1)
                rejected_logratios_sorted = torchsort.soft_sort(rejected_logratios,
                    regularization_strength=0.1)
            delta_sorted = chosen_logratios_sorted - rejected_logratios_sorted
         if self.AOT_loss == "hinge":
        losses = torch.relu(self.beta - delta_sorted)**2
    elif self.AOT_loss == "logistic":
        losses = (
                -F.logsigmoid(self.beta *delta_sorted) * (1 - self.label_smoothing)
                - F.logsigmoid(-self.beta * delta_sorted) * self.label_smoothing
                )
```

```
elif self.loss_type == "AOT_pair":
     pi_logratios = policy_chosen_logps - policy_rejected_logps
     ref_logratios = reference_chosen_logps - reference_rejected_logps
    if self.sort_type == "hard_sort":
        pi_logratios_sorted, _ =  torch.sort(pi_logratios, dim=0)
        ref_logratios_sorted, _ =  torch.sort(ref_logratios, dim=0)
    elif self.sort_type == "soft_sort":
        pi_logratios_sorted = torchsort.soft_sort(pi_logratios, regularization_strength=0.1)
                ref_logratios_sorted = torchsort.soft_sort(ref_logratios, regularization_strength
                    =0.1)
            delta_sorted = pi_logratios_sorted - ref_logratios_sorted
         if self.AOT_loss == "hinge":
        losses = torch.relu(self.beta - delta_sorted)**2
    elif self.AOT_loss == "logistic":
        losses = (
                -F.logsigmoid(self.beta *delta_sorted) * (1 - self.label_smoothing)
                - F.logsigmoid(-self.beta * delta_sorted) * self.label_smoothing
                )
```

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Parametric Rate for AOT

Theorem (Informal)

Under appropriate Assumptions we have:

$$\mathbb{E}\ \mathrm{OT}_h\left((r\circ\pi_{\hat{\theta}_n})_\sharp\mu_+,(r\circ\pi_{\hat{\theta}_n})_\sharp\mu_-\right)-\mathrm{OT}_h\left((r\circ\pi_{\theta^*})_\sharp\mu_+,(r\circ\pi_{\theta^*})_\sharp\mu_-\right)\lesssim n^{-\frac{1}{2}},$$

where $\lesssim$ refers to inequality up to constants that depend only on constants in the assumptions.

The model estimated from emprical samples with AOT generalizes on unseen samples with a rate $n^{-\frac{1}{2}}$

Elements of the Proof 1/2

$$\min_{\pi_\theta \in \mathcal{H}} \int_0^1 h\left(Q_{(r\circ\pi_\theta)_\sharp \mu_+}(t) - Q_{(r\circ\pi_\theta)_\sharp \mu_-}(t)\right) dt = \min_{\pi_\theta \in \mathcal{H}} \mathrm{OT}_h\left((r\circ\pi_\theta)_\sharp \mu_+, (r\circ\pi_\theta)_\sharp \mu_-\right) \tag{uAOT$_h$}$$

Define the OT cost $c : [-M, M] \times [-M, M] \to [0, R]$ such that $c(z, z') = h(z - z')$, for $z, z \in [-M, M]$. Define the c-transform of a function $\varphi : [-M, M] \to \mathbb{R}$:

$$\varphi^c(z) = \inf_{z' \in [-M,M]} h(z - z') - \varphi(z).$$

In our setting, a function is called c-concave if there exists $\psi : [-M, M] \to \mathbb{R}$ such that $\varphi = \psi^c$. Define:

$$\mathcal{F}_c = \{\varphi : [-M, M] \to [-R, R], \varphi \text{ is } c\text{-concave, with } ||\varphi^c||_\infty \leq R\}$$

By duality (Theorem 5.10 in [Villani(2009)]) we have:

$$\mathrm{OT}_h\left((r\circ\pi_\theta)_\sharp \mu_+, (r\circ\pi_\theta)_\sharp \mu_-\right) = \sup_{\varphi \in \mathcal{F}_c} \int \varphi(r\circ\pi_\theta) d\mu_+ - \int \varphi^c(r\circ\pi_\theta) d\mu_-.$$

Elements of the proof 2/2

Replacing the dual expression of OT_h in (uAOT_h), we see that (uAOT_h) can be cast as a min-max problem:

$$\min_{\pi_\theta \in \mathcal{H}} \sup_{\varphi \in \mathcal{F}_c} \int \varphi(r \circ \pi_\theta) d\mu_+ - \int \varphi^c(r \circ \pi_\theta) d\mu_-. \tag{6}$$

Given samples $\hat{\mu}_+^n = \frac{1}{n}\sum_{i=1}^n \delta_{(x_{i,+},y_{i,+})}$ and $\hat{\mu}_-^n = \frac{1}{n}\sum_{i=1}^n \delta_{(x_{i,-},y_{i,-})}$, the empirical problem is:

$$\min_{\pi_\theta \in \mathcal{H}} \sup_{\varphi \in \mathcal{F}_c} \int \varphi(r \circ \pi_\theta) d\hat{\mu}_+^n - \int \varphi^c(r \circ \pi_\theta) d\hat{\mu}_-^n. \tag{7}$$

Let π_{θ^*} be the population minimizer of (uAOT_h) and $\pi_{\hat{\theta}_n}$ be the solution of the empirical problem (7). We have the following sample complexity bound for the violation of stochastic dominance in AOT unpaired:

$$\mathbb{E}\ \mathrm{OT}_h\left((r \circ \pi_{\hat{\theta}_n})_\sharp \mu_+, (r \circ \pi_{\hat{\theta}_n})_\sharp \mu_-\right) \leq \underbrace{\mathrm{OT}_h\left((r \circ \pi_{\theta^*})_\sharp \mu_+, (r \circ \pi_{\theta^*})_\sharp \mu_-\right)}_{\text{Optimal Almost FSD Violation}}$$

$$+ \underbrace{2\mathcal{R}_n(\mathcal{F}_c; (r \circ \pi_{\theta^*})_\sharp \mu_+) + 2\mathcal{R}_n(\mathcal{F}_c^c; (r \circ \pi_{\theta^*})_\sharp \mu_-)}_{\text{One dimensional OT sample complexity with optimal } \theta^*}$$

$$+ \underbrace{2\mathcal{R}_n(\mathcal{F}_c \circ r \circ \mathcal{H}; \mu_+) + 2\mathcal{R}_n(\mathcal{F}_c^c \circ r \circ \mathcal{H}; \mu_-)}_{\text{Complexity of learning in } \mathcal{H} \text{ via the 1D OT problem}},$$

where $\mathcal{R}_n(\mathcal{F}; \nu) = \mathbb{E}\sup_{\varphi \in \mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^n \sigma_i \varphi(Z_i)\right|$ is the Rademacher Complexity and for $i = 1 \ldots n$, σ_i are independent Rademacher random variables and $Z_i \sim \nu$ iid.

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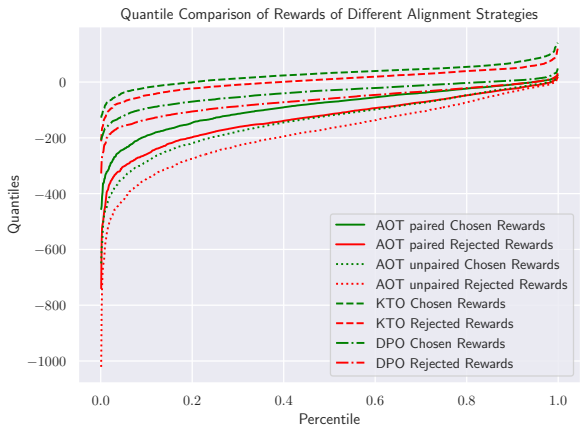
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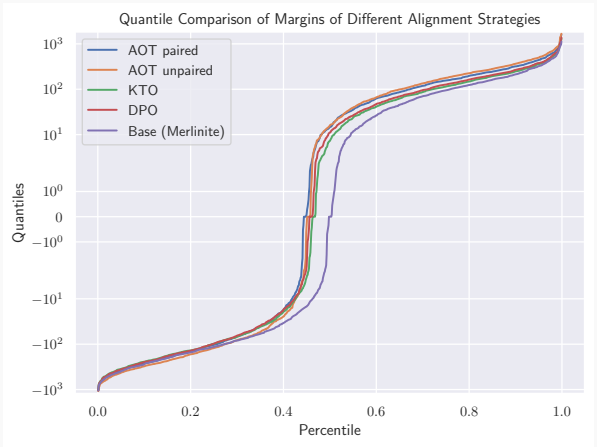
Statistical Properties

Results

Distributional Preference



(a) Stochastic Dominance of Reward of Chosen on Rejected: AOT achieves a larger margin between the quantile plots of chosen and rejected rewards.



(b) Stochastic Dominance of AOT's optimized policy margin (between Chosen on Rejected) on the margin of the reference policy.

Figure 1: AOT in the paired & unpaired settings enables first-order stochastic dominance of the chosen reward distribution on the rejected distribution (a). The margin between the quantiles of chosen and rejected rewards is larger than alternative strategies. In (b), we see that AOT's policy chosen to rejected log-likelihood ratio dominates that ratio for the base model and alternative strategies.

Results on Merlinite 7B Alignment

	AlpacaEval (GPT4)	ARC	Hellaswag	MMLU	Truthful	Winogrande	GSM8K
AOT paired	29.9	82.5	66.1	62.9	50.8	74.4	53.1
AOT unpaired	**31.3**	82.5	**66.2**	62.8	**51.1**	74.4	51.8
DPO	27.4	**82.8**	65.8	**63.1**	50.6	74.3	52.0
KTO	24.9	82.7	65.4	63.0	48.7	**74.9**	**53.9**
IPO	27.7	82.4	65.1	63.0	46.5	74.0	52.3
Merlinite-7B	17.1	81.6	63.2	62.6	42.0	73.9	45.2

Table 1: `Merlinite-7B` trained on UltraFeedback Binarized. AOT results in the best performing LLM as compared to the alternative alignment algorithms on AlpacaEval, and is competitive across the other benchmarks that are evaluated in the zero shot regime.

Ablations on AOT Losses and Batch size

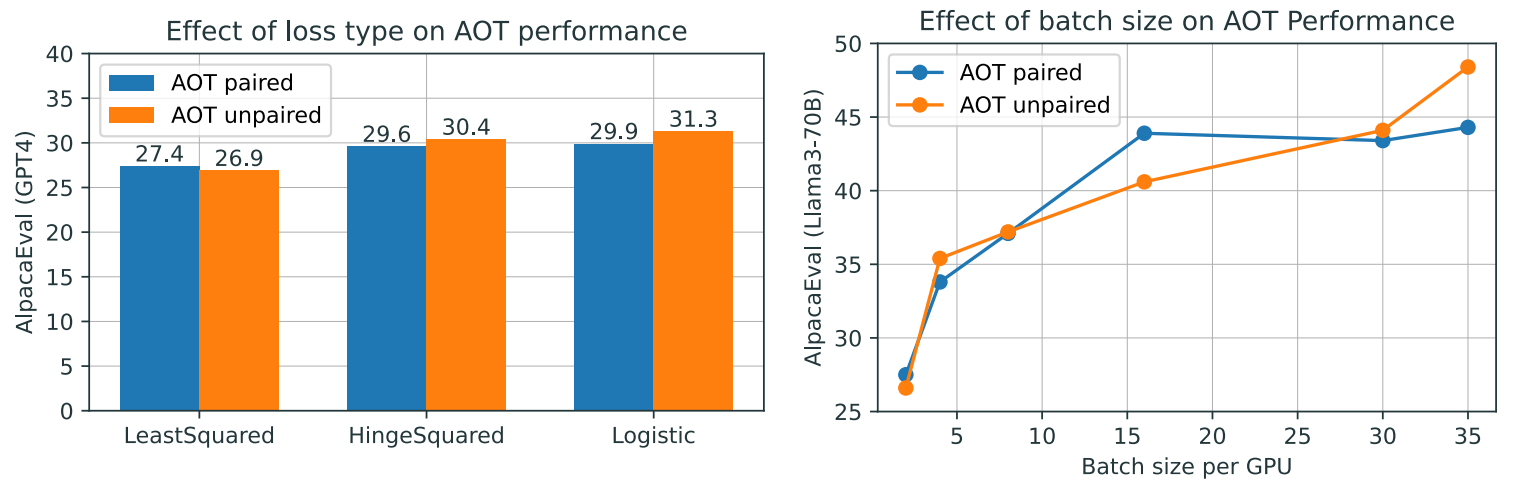


Figure 2: Impact of batch size and loss type on AOT performance. The batch size is the effective number of samples in the mini-batch per GPU. We found the logistic loss to be performing better than least squared or hinge squared losses (all using $\beta = 0.01$). As we increase batch size, we also observed improvement in AOT performance, which is expected as more samples per minibatch results in a better effect of stochastic dominance (conforming Corollary 5).

Ablation Study on β

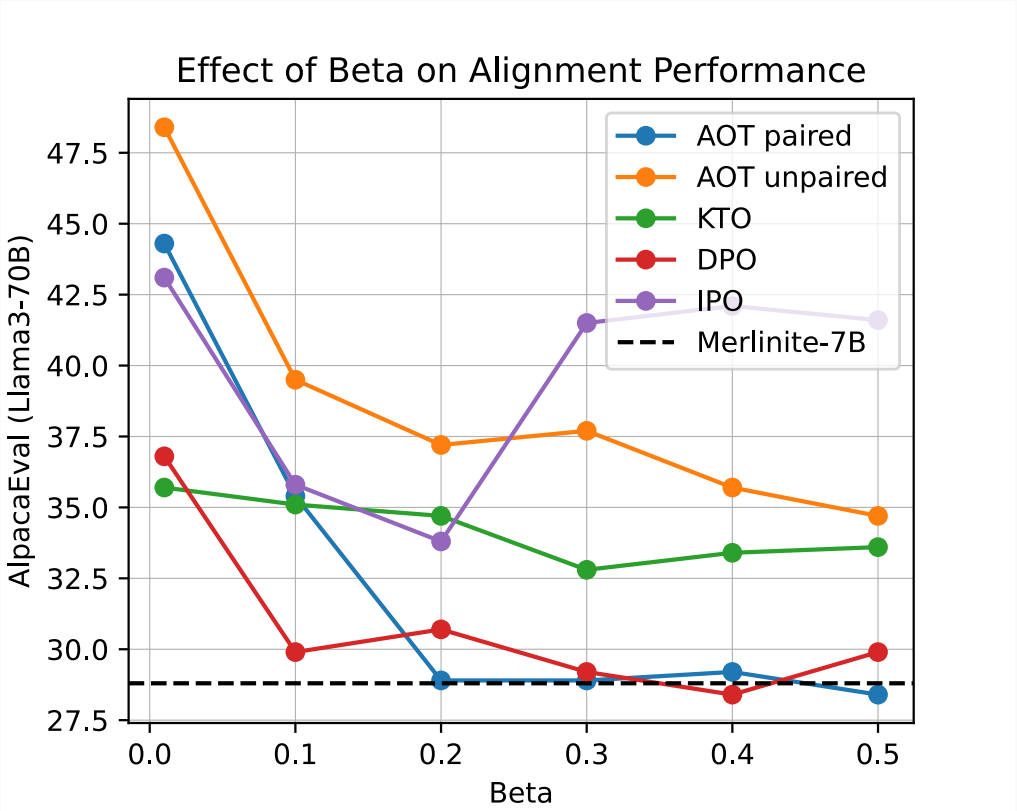


Figure 3: Impact of (β) parameter on performance of different alignment algorithms. β controls the divergence of the policy model from the initial reference model (low beta - more divergence, high beta - less divergence). We see a general trend that with higher betas, LLMs alignment decreases the performance. Hence, for all experiments, we selected $\beta = 0.01$ as a default value.

Outline

Pointwise Preference Optimization

From Pointwise to Distributional Preference

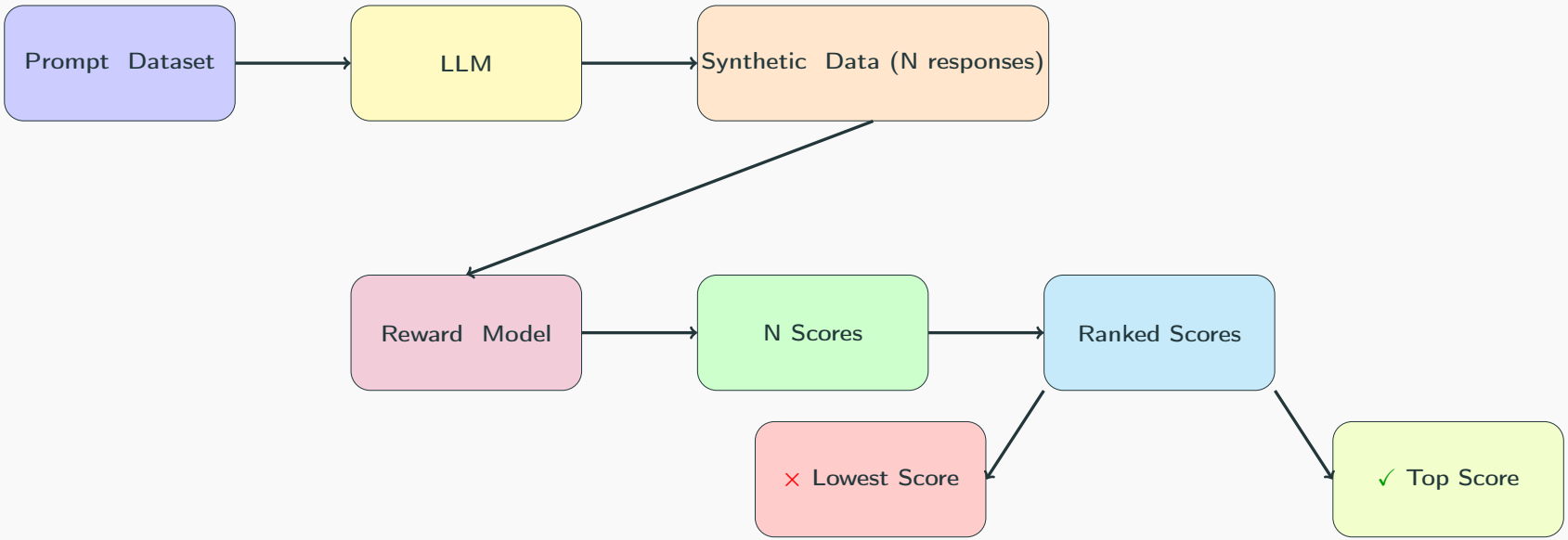
Distributional Preference Relaxation to a 1D Optimal Transport Problem

Statistical Properties

Results

Recipe: Synthetic Data+ Reward Models + AOT

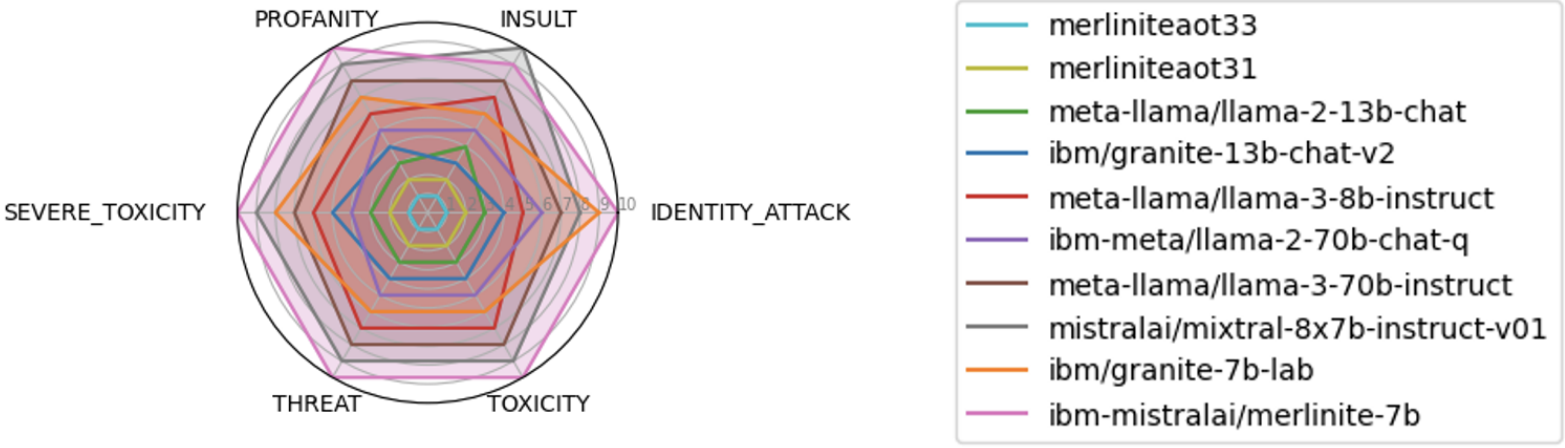
On-Policy Synthetic Data Scored with a Reward Model as Preference Data for AOT



- LLM is Merlinite 7B
- Reward Model is Pair RM
- Prompt Dataset: Ultrachat 200K
- Benchmarking dataset: AttaQ dataset that measures risks of LLMs
- Toxicity Metrics from Perspective API

Toxicity Evaluation on AttaQ dataset with Perspective API

Rankings based on our method:



Rankings based on average scores:



Figure 4: Toxicity Evaluation: AOT33 trained on synthetic data with AOT

Thank You !

mroueh@us.ibm.com

Code in TRL: ▸ HuggingFace Link

Paper on arxiv: `https://arxiv.org/abs/2406.05882`

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Multivariate Stochastic Dominance via Optimal Transport and Applications to Models Benchmarking

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Kristjan Greenewald† **Youssef Mroueh†**

NeurIPS 2024

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Stochastic Dominance

Definition

$$X \underset{\text{FSD}}{\succcurlyeq} Y \text{ if and only if } \mathbb{P}(X \leq t) \leq \mathbb{P}(Y \leq t) \; \forall t \in \mathbb{R}$$

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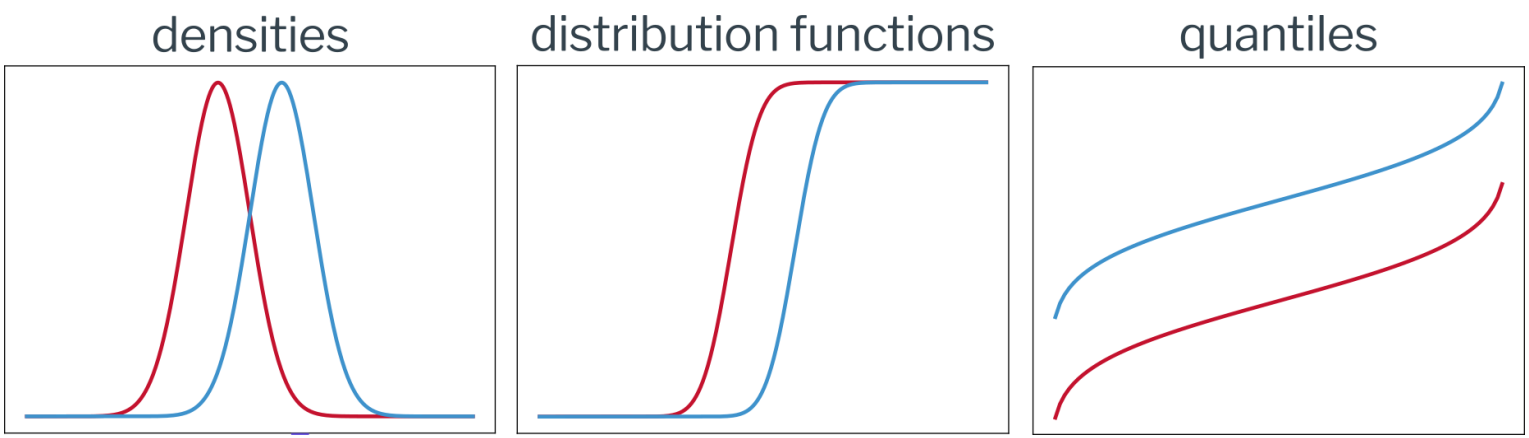
Widely used in economics and statistics to compare random outcomes by preference

Stochastic Dominance

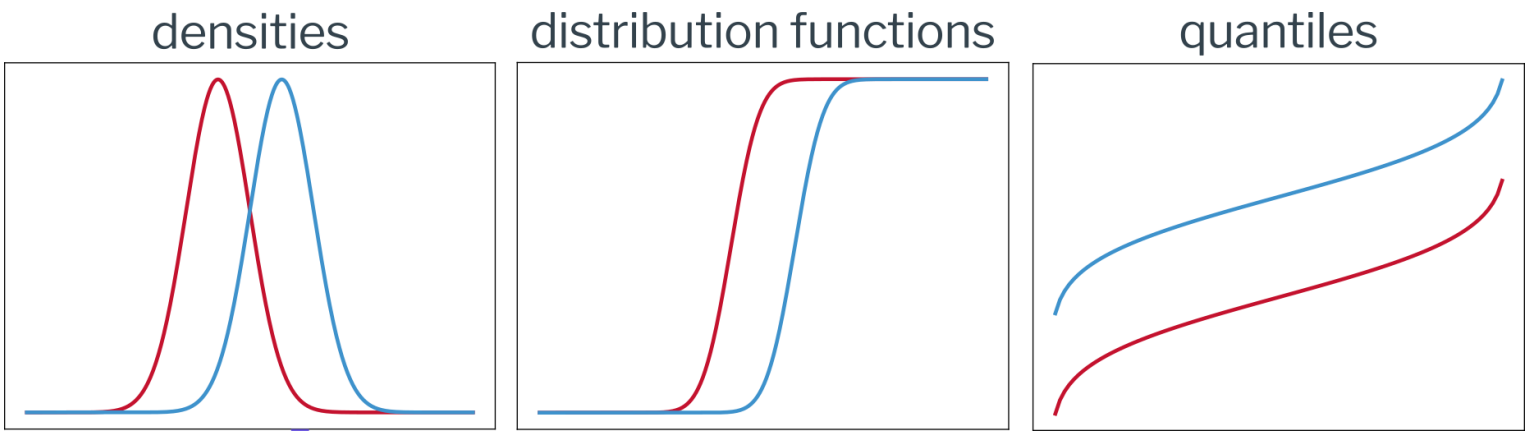
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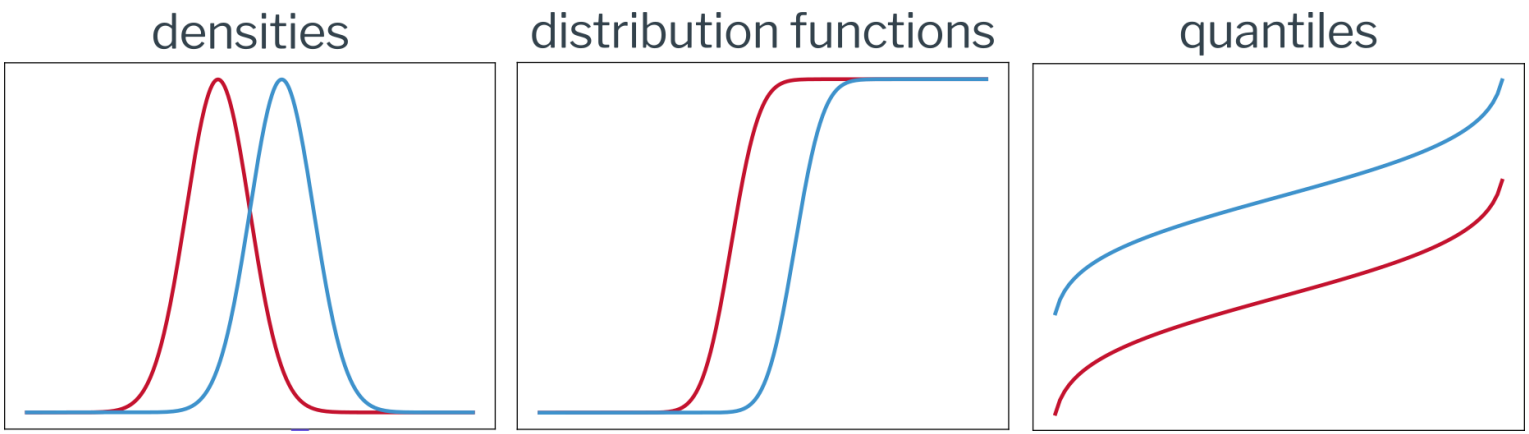


Stochastic Dominance and OT



$X \underset{\text{FSD}}{\succcurlyeq} Y$ equivalent to $F_Y^{-1}(t) \leq F_X^{-1}(t)\ \forall t \in (0,1)$ so

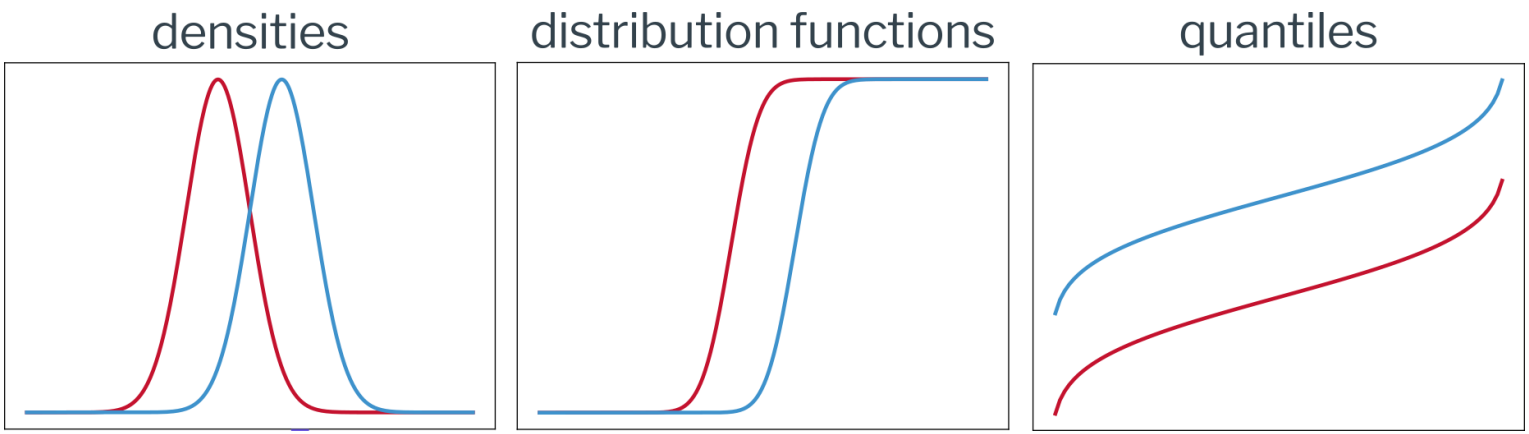
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$$0 = \int_0^1 \max\left\{F_Y^{-1}(t) - F_X^{-1}(t), 0\right\}^2 dt$$

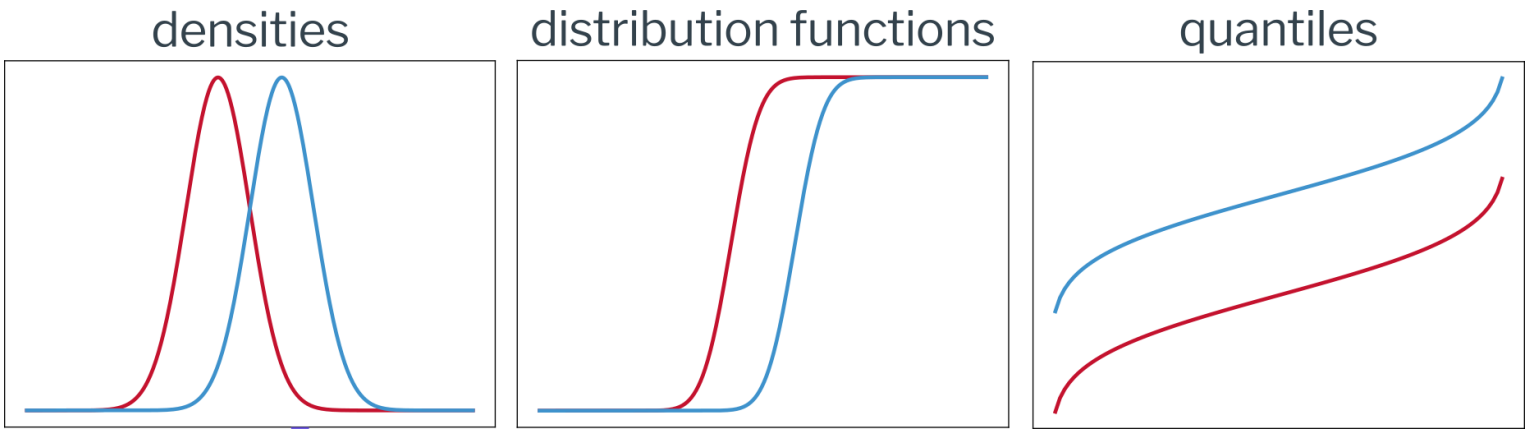
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$$= \underbrace{\inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int \max\{y - x, 0\}^2 \, d\pi(x,y) \right\}}_{\mathsf{OT}_c(\mu,\nu),\ c(x,y)=\max\{y-x,0\}^2}$$

Multivariate Stochastic Dominance

Let $h : \mathbb{R} \to [0, \infty)$ such that $h(z) = 0$ if and only if $z \in (-\infty, 0)$, then can define $X \underset{\text{FSD}}{\succcurlyeq} Y$ for d-dimensional vectors if and only if $\mathsf{OT}_c(\mu, \nu) = 0$ for $c(x, y) = \sum_{i=1}^{d} h(y_i - x_i)$.

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Issue

If $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$,

$$\mathbb{E}\left[\left|\mathsf{OT}_c(\hat{\mu}_n, \hat{\nu}_n) - \mathsf{OT}_c(\mu, \nu)\right|\right] \lesssim n^{-\frac{1}{d}}$$

Entropic Regularization

Replace $\mathsf{OT}_c(\mu, \nu)$ with

$$\mathsf{OT}_{c,\lambda}(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int c d\pi + \lambda \mathsf{D}_{\mathrm{KL}}(\pi \| \mu \otimes \nu) \right\},$$

computationally and statistically efficient!

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If $c(x, y) = \sum_{i=1}^{d} h(y_i - x_i)$ and h is smooth and Lipschitz continuous, let $\check{c}(x, y) = c(y, x)$ and define

$$\varepsilon_{h,\lambda}(\mu, \nu) = \frac{\mathsf{OT}_{c,\lambda}(\mu, \nu)}{\mathsf{OT}_{c+\check{c},\lambda}(\mu, \nu)} \in [0, 1]$$

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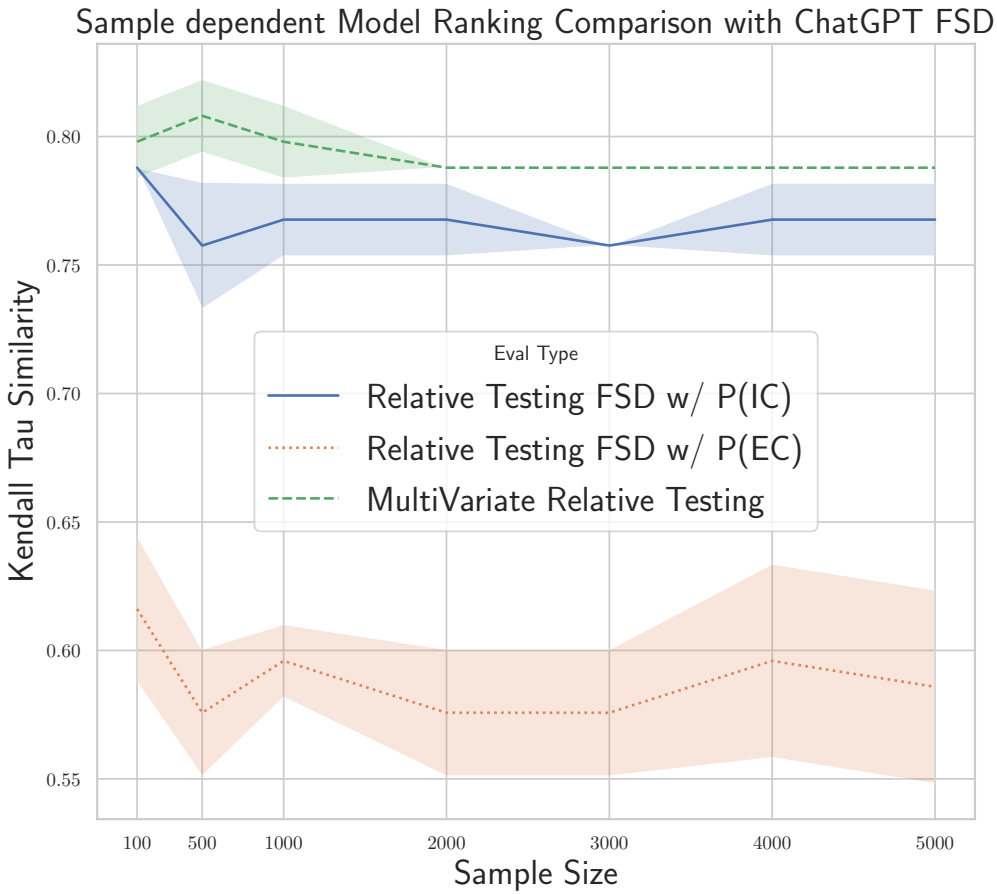
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If $\varepsilon_{h,\lambda}(\mu, \nu) \approx 0$, $X \underset{\mathrm{FSD}}{\succsim} Y$ whereas if $\varepsilon_{h,\lambda}(\mu, \nu) \approx 1$, $Y \underset{\mathrm{FSD}}{\succsim} X$.

Numerical Experiments



Using $\varepsilon_{h,\lambda}$ to generate a ranking of 12 LLMS based on automatic metrics. Kendall Tau similarity is taken with respect to ChatGPT scoring of the top model. Multivariate FSD is seen to outperform portfolio aggregation with independent and empirical copulas.